MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY
(BICT / BSCEICT/ BSCEICTU / BAE / BLIS)

END OF SEMESTER EXAMINATION

ICT 4703 NETWORK ADMINISTRATION & INFORMATION SECURITY

TIME ALLOWED: 2:30 HOURS **DATE : 30th MARCH , 2017**

INSTRUCTIONS

i. This paper contains **Two** pages. Please check.
ii. There are **FourQuestions**in this paper. Please check.
iii. Answer **ALL** questions.
iv. Marks for each question are indicated.

Question 1

a. Explain OSI architecture for security. **[6 Marks]**

b. What are the different types of security services. **[10Marks]**

c. Give counter measures for Confidentiality, Integrity, Authentication and Denial of Service for web security threats. **[8 Marks]**

Question 2

a. With the help of a diagram, explain CBC Operation Mode for block ciphers. **[8 Marks]**

b. Cryptography can be used for many things. It can be used to encrypt data to be archived on a hard disk for many years, or it can be used to protect a critical message being sent across the internet. Is the key selection issue the same for these cases? **[2 Marks]**

c. Khalidwe and Bright are having a debate about Public Key Infrastructure. Khalidwe says that it is simply a way of authenticating users. However, Bright insists that it is a type of encryption algorithm. You have been asked to help them make a decision.

i. In C above, whose opinion is correct or incorrect? **[2 Marks]**

ii. Provide **one** reason why it is either correct or not. **[2 Marks]**

iii. Briefly, give an outline of the purpose of PKI. **[3 Marks]**

iv. Explain what is meant by a Certificate Authority and Digital Certificate. **[2 Marks]**

d. Explain briefly what a MAC is. **[5 Marks]**

e. Explain how substitution techniques for cryptography work. **[2 Marks]**

f. What will be the cipher text of plaintext "attack from south east" with "point" as key using Vigenere Cipher? **[10Marks]**

Question 3

a. Confidentiality, Integrity and Availability are core attributes of Information Security. Identify **three** threats to Wireless Network that could compromise security. **[3 Marks]**

b. State the security attribute that is compromised by each threat. **[3 Marks]**

c. What is a firewall? **[1 Mark]**

d. Explain the different types of firewalls. **[8 Marks]**

e. In SSL Handshake protocol, there is also a **Heartbeat Extension to the SSL/LTS.** Explain how the Heartbeat Extension works. **[6 Marks]**

f. Explain the operation of PGP with confidentiality service. **[5 Marks]**

g. Describe the use of S/MIME. **[5 Marks]**

Question 4

a. What is the difference between a virus and a worm **[2 Marks]**

b. TCP/IP protocol stack provides confidentiality and authentication for information security in all its layers. Mention the security provided in each layer. **[4 Marks]**

c. You receive the following email from the Mzuni IT Section:

Dear Mzuni Student,

Please be informed that next week we will be deleting all email activities that are not active in order to create space for more users. Send us the following information in order to allow us maintain your email account to avoid inconveniences.

** Full Name*

** Email login and Password:*

** Alternate email:*

If you have any question, contact us. Thank you.

i. What should you do? **[1 Mark]**
ii. What can be the risk of not doing what you did? **[1 Mark]**

END OF QUESTION PAPER